Example 1.11

Add : $\frac{-5}{9}, \frac{-4}{3}, \frac{7}{12}$

Solution:

LCM of $9,3,12 = 36$

So, $\frac{-5}{9}+\frac{-4}{3}+\frac{6}{12}=\frac{-5}{9}\times\frac{4}{4}+\frac{-4}{3}\times\frac{12}{12}+\frac{7}{12}\times\frac{3}{3}$

$$=\frac{-20}{36}+\frac{-48}{36}+\frac{21}{36}=\frac{-20-48+21}{36}=\frac{-47}{36}$$

1.3.2 Additive Inverse

The additive inverse of a rational number is another rational number which when added to the given number, gives zero.

For example, $\frac{4}{3}$ and $\frac{-4}{3}$ are additive inverses of each other, since their sum is zero.

Think



Is zero a rational number? If so, what is its additive inverse?

1.3.3 Subtraction

Subtraction is simply adding the additive inverse.

Example 1.12

Subtract : $\frac{9}{17}$ from $\frac{-12}{17}$

Solution:

Now, $\frac{-12}{17}-\frac{9}{17}=\frac{-12}{17}+\left(\frac{-9}{17}\right)=\frac{-12-9}{17}=\frac{-21}{17}$

Example 1.13

Subtract : $\left(-2\frac{6}{11}\right)$ from $\left(-4\frac{5}{22}\right)$

Solution:

Now, $\left(-4\frac{5}{22}\right)-\left(-2\frac{6}{11}\right)$

$$=\frac{-93}{22}-\left(\frac{-28}{11}\right)$$

$$=\frac{-93}{22}+\frac{28}{11}=\frac{-93+28\times2}{22}$$

$$=\frac{-93+56}{22}=\frac{-37}{22}=-1\frac{15}{22}$$

1.3.4 Multiplication

Product of two or more rational numbers is found by multiplying the corresponding numerators and denominators of the numbers and then writing them in the standard form.

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